The real value of the nominal interest payment will amount to R20 000 each year. The nominal redemption value of R1 191 016 at the end of 20x7 will have a real value of R1m [1191016/1.191]. If the inflation rate changes in 20x7, to say 5%, then the nominal interest payment in 20x7 would reduce to R23 596 and the nominal redemption value would fall to R1 179 780.[6]

Assume that it is estimated that the future inflation rate is 6%. The company could have issued straight bonds with a quoted coupon rate of 8.12%. Assume that the actual inflation rate turns out to be 6% each year. Then, if we issued straight bonds or inflation-linked bonds, the YTM for both bonds would be the same.

Inflation-linked	20x4	20x5	20x6	20x7
Principal	(1,000,000)	21,200	22,472	1,191,016
Coupon pmt				23,820
	(1,000,000)	21,200	22,472	1,214,836
YTM (IRR)	8.120%			
Straight	**20x4**	**20x5**	**20x6**	**20x7**
Principal	(1,000,000)	81,200	81,200	1,000,000
Coupon pmt				81,200
	(1,000,000)	81,200	81,200	1,081,200
YTM (IRR)	8.120%			

However, we will now assume that right after the issue of the bonds, the inflation rate increases to 10% per year. The straight bonds will continue to have a yield of 8.12%, which is lower than the inflation rate (and which will result in a fall in the value of the bonds) while the YTM of the inflation-linked bonds will increase to 12.2%.

Inflation-linked	20x4	20x5	20x6	20x7
Principal	(1,000,000)			1,331,000
Coupon pmt		22,000	24,200	26,620
	(1,000,000)	22,000	24,200	1,357,620
YTM (IRR)	12.200%			

For investors, a government bond while risk-free in relation to the repayment of capital, carries significant inflation risk while a government inflation-linked bond protects investors from the risk that higher inflation will reduce the real value of bonds and the real value of the bond's cash flows. An inflation-linked government bond reflects more closely what we call a riskless asset. Some companies are able to increase prices with inflation and are therefore effectively hedged if they issue inflation-linked bonds. Pension funds will often have inflation-linked liabilities and may wish to protect the real value of their investments by investing in inflation-linked

bonds. The South African government and Eskom have made a number of inflation-linked bond issues such as the R212, R202 and R210. The terms of inflation-linked bonds will normally protect against deflation so that the principal amount will never be less than the original amount. Page 13-36

8 ALTERNATIVE SOURCES OF FINANCE

Technology and the Internet is beginning to have a significant impact on the development of alternative sources of finance. This largely involves bringing lenders and borrowers together by creating online platforms for borrowing and lending without banks performing their traditional intermediary role. The role of traditional lending and borrowing has remained essentially unchanged since the Middle Ages, with banks performing both the risk assessment and the borrowing and lending functions. Yet credit ratings have enabled large firms to issue bonds and notes directly to debt investors such as pension funds, insurance firms and asset management companies. It has also allowed them to list bonds on securities exchanges. The new alternative online financing structures will mainly enable smaller companies and individuals to access loan and equity finance directly from investors.

Peer-to-peer lending and SME online lending platforms

Peer-to-peer lending grew significantly in the USA and the UK until 2022. It represents borrowing and lending by unrelated firms or individuals in an online marketplace where investors and borrowers negotiate interest rates. The firm providing the online marketplace will undertake credit checks on borrowers. While most peer-to-peer loans are to individuals, we will focus on peer-to-peer lending to business firms. Interest rates are set by reverse auction whereby lenders bid on the interest rates charged to firms. So, for example, assume that a small company, XYZ (Pty) Ltd, wishes to borrow R1m repayable over 5 years. It will go onto the online website to register and to lodge its funding request for R1m. The firm providing the online marketplace will undertake a credit check on the firm and if the credit risk assessment is positive, then the request for funding will be listed on the website. Investors (lenders) will then offer to lend at specified interest rates. Assume that the following investors (lenders) have bid to lend the following amounts, at the indicated rates.

Lender	Rate	Amount
S. Buthelezi	9.5%	500,000
P. Smit	11.0%	500,000
S. Patel	12.0%	500,000

XYZ (Pty) Ltd will then choose to borrow from S. Buthelezi and P. Smit who will forward the funds via the online marketplace to the company, which will then often be required to start making repayments in the form of monthly instalments, inclusive of the agreed interest rate over the next 5 years.

In South Africa, a number of firms such as **Funderjet** provide a peer-to-peer lending platform, but others have left the sector. We are going to focus on the UK firm, **Funding Circle,** which is one of the world's leading online marketplaces for small businesses and which had 77 000 retail investors (lenders) in the UK in 2018. At the time, Funding Circle had facilitated the lending of £4.2 billion (which is about R100bn) to 42 000 small businesses in the UK. Yet, you know the sector is in trouble when Funding Circle, one of the leading peer-to-peer lending platforms, decides to change its business model. Why? A number of peer-to-lending platforms such as Lendy and FundingSecure collapsed due to poor lending decisions, increased rates of default, sector concentration, and a lack of discipline, thereby leaving investors experiencing significant losses. Lendy collapsed with 9000 loan investors owed almost £160m. In China and the USA there were other high profile failures. This attracted increased regulation of the sector and Funding Circle changed its business model in March 2022 by closing its doors to retail investors (lenders). It continues to lend to SMEs using its online platform but only derives funds from institutional investors. We wrote the following in 2018:

> Peer-to-peer lending has the potential to turn banking upside down or at least create competition. Investors will receive higher returns and companies may borrow at lower rates. Page 13-37
>
> However, at the end of the day, the success of peer-to-peer lending will be determined by the effectiveness and the disciplined use of appropriate credit scoring models. As we saw with the demise of African Bank, the effective risk assessment of borrowers is critical.

Once investors lose money even with only one peer-to-peer lender, the sector will lose retail investors. By 2023, Funding Circle had 144 institutional investors and had made 135000 loans to SMEs totalling £15.2bn. A loan decision could be made within an hour and interest rates were from 8.9% in 2023. Loans are from £10000 to £500000 and for 6 months to 6 years. Funding Circle is one of the largest platforms for business

loans in the world. The platform offers institutions such as asset management firms and other institutions access to small business loans as an asset class. Although Funding Circle has moved on from peer-to-peer lending, the technology, systems and credit risk scoring models it employed is being employed to make lending decisions to SMEs. This creates competition for the banks and provides SMEs with finance at a lower cost with a lower administrative burden. In South Africa, the firm *SME South Africa* is playing a role of partnering with finance providers to make business loans to SMEs as well as providing other services.

Crowdfunding

Crowdfunding is similar to peer-to-peer lending, but mostly relates to the funding by individuals of start-up firms or projects. For example, filmmakers may use crowdfunding from individuals who believe in their projects in order to complete the shooting and production of a film. Equity-based crowdfunding involves investors using an online platform to acquire equity in a firm. **Crowdcube** is a leading investment crowdfunding platform in the UK, which has facilitated the successful funding of over 1300 businesses at over £1bn (about R23bn) and has over 1.5m registered investors. Crowdcube enables individuals to acquire stakes in businesses alongside professional investors. Investors choose which business to invest in, in exchange for an equity stake, which will often be in start-up, early stage or growth companies. A person can invest from as little as £10 in a company. Crowdcube requires a firm to submit a "pitch", which includes a business plan and financials, which are then published. Once the firm hits its target, then the deal is completed. Crowdcube warns investors as follows:

> Investing in start-ups and early-stage businesses involves risks, including illiquidity, lack of dividends, loss of investment and dilution, and it should be done only as part of a diversified portfolio.

In the USA, there are three main crowd funding platforms; Kickstarter (>$7.5bn), Indiegogo and CircleUP. In South Africa, crowd funding platforms include Thundafund, Jumpstarter, Backabuddy and Uprise.Africa. Thundafund reports raising close to $5.6m (R108m) from 26 488 supporters to fund 1 747 projects by October 2023. Projects include 3D printing, conservation and climate change, books, Rooibos products, kids' health, toys, water purification, farming equipment, a music album, a film documentary, gin and brandy, and coffee.

Online invoice trading platforms

An online invoice-trading platform is a form of asset-based financing and enables companies to sell outstanding invoices to investors for a fee on an online marketplace. A company registers on the online exchange, and then selects invoices, against which it wishes to raise finance. Investors bid against one another and the winning bids are selected and funds are advanced to the company. When the customer pays the invoice, then investors are repaid the money advanced, plus a fee. However, in most cases we have firms that employ online portals and specialise in invoice discounting but partner with a financial institution or bank to provide Page 13-38 the financing rather than obtain financing from retail investors. Pick n Pay and its partner Addendum provide an invoice discounting service for Pick n Pay's suppliers (see Chapter 12). A bank such as FNB provides an invoice discounting service to provide firms with short-term financing. FNB also provides firms with financing to enable firms to take advantage of early payment discounts. This represents a short-term source of finance for companies.

Supply chain finance

Supply chain finance refers to a situation in which large customers can support their small suppliers as large customers will usually have a lower cost of capital. The large customers can provide direct loans to their suppliers. Alternatively, the smaller suppliers could sell invoices to banks, which then rely on the higher credit quality of the customers rather than the credit quality of the supplier, thereby resulting in a lower discount or fee. Pick n Pay is currently doing this for its suppliers.

9 FINANCING FOR BLACK ECONOMIC EMPOWERMENT (BEE) ENTITIES

The economic empowerment of black South Africans is a business imperative. Black Economic Empowerment (BEE) legislation has been enacted and there are agreements between government and industry sectors (industry charters) with empowerment targets relating to ownership, management control, skills development, employment equity and procurement. Limited access to financing by BEE entities has created challenges to the structuring of the BEE ownership of companies. It has also resulted in complex and highly leveraged structures, some of which have failed and others of which have become highly successful as equity markets have rebounded. Whilst BEE

is a wide-ranging topic, we will focus on BEE equity structures and sources of finance.

Firstly, in relation to BEE financing, entities such as the IDC, Business Partners and the Public Investment Corporation (PIC) are increasingly financing BEE transactions. The investment banks have also specialised in advising and sourcing financing for BEE entities. Banks now have divisions which are providing expertise on the implementation of the BEE charters, the provision of tax efficient structures and advising on equity and hybrid instruments such as convertible preference shares and options. There is also vendor financing. The Companies Act permits a company to provide financial assistance for the acquisition of its own shares in order to facilitate BEE ownership. The shares may be issued to the BEE entity at a discount to the current share price (normally 10%) so that the current shareholders provide assistance through value dilution rather than the company providing financial assistance. The source of financing for BEE entities with limited funds has included the following alternatives:

- *Vendor financing in the form of loans or preference shares*. The vendor (selling company) provides a subordinated loan to the BEE entity to acquire the shares in the company. Also, the vendor may subscribe for preference shares in the BEE entity so that the BEE entity can use the proceeds from the preference share issue to purchase the ordinary shares in the company. The vendor can also use hybrid securities such as convertible preference shares, convertible debentures or options to enable the BEE entity to acquire shares in the vendor.
- *Loans or debentures*. Banks may provide loan finance against the security of the underlying shares. The highly leveraged structure means that this is facilitated if the BEE entity acquires the shares at a discount to the prevailing share price. Otherwise the BEE entity may issue debentures. Investment banks may take a small equity stake in the BEE entity. The disadvantage of using loan finance is that interest is not deductible for the acquisition of shares. The acquisition of operating assets will, however, result in the deductibility of interest if BEE is structured in terms of a joint venture in a project or if the business is sold to a new company (Newco) in which the BEE partner acquires equity. The loan finance is provided to Newco.

- *Preference shares issued to banks*. In this case, a bank will subscribe for preference shares Page 13-39 issued by the BEE entity. This will be issued at dividend rate which is linked to the prime lending rate, normally at 70–74% of the prime rate. The advantage here is that the banks receive tax-free dividend income whilst the BEE entity pays a lower financing rate than the interest rate. The bank is exempt from the dividend withholding tax (DWT) as it is a resident South African company.
- *Asset securitisation or initial public offering (IPO)*. A BEE company which has equity in existing investments may undertake an IPO and list the company to raise funds to finance further BEE transactions. However, this may result in a dilution of black ownership. The BEE entity may also transfer some of its assets to a separate company and issue securities against those assets to raise funds. Again the BEE entity is required to have already a significant equity interest in its assets to enable it to undertake securitisation.

BEE transactions will require that dividends due to the BEE entity from the investment in ordinary shares (or convertible preference shares) are first used to pay off the interest or preference dividends or capital due to the financiers of the BEE transaction.

Let's evaluate a BEE structure. Let's assume that there is an operating company, Metro Ltd. The company is 100% owned by the shareholder, M Ltd. The company has decided to sell 51% of the ordinary shares in Metro Ltd at a price of 10% below current market value. Let's assume that the consideration amounts to R540 million or R7.00 per share (total value is R1 058,82m).

BEE Holdings Ltd forms a new company, a 100% owned subsidiary (let's call it SPV Ltd) and subscribes for the ordinary equity of SPV Ltd for a total consideration of R30 million. SPV Ltd then issues preference shares to a bank paying a dividend rate of 74% of the prime rate, thereby raising a further R510m. SPV Ltd uses these funds to acquire the ordinary shares in Metro Ltd. Let's assume that the ordinary shares have just paid a total ordinary dividend of R41m for the last financial year and ordinary dividends are expected to grow at a rate of 8% per year. Figure 13.3 depicts the proposed structure.

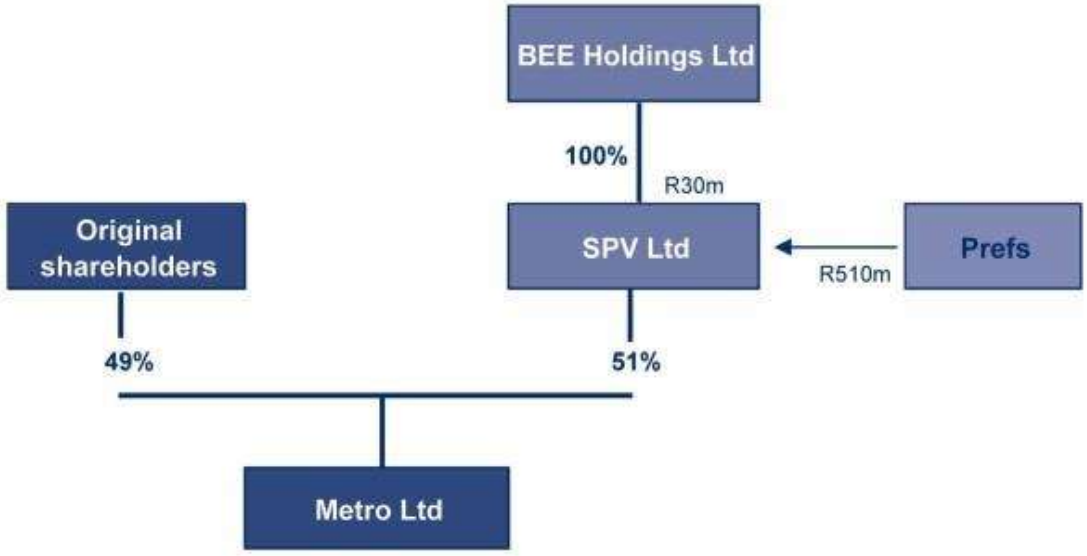


Figure 13.3 BEE financing structure

Page 13-40 The preference shares are redeemable in 5 years' time. Let's assume that over the 5-year period, the value of ordinary shares more than doubles (let's assume a 17.8% increase in the share price per year and the market capitalisation of Metro Ltd in 5 years is about R2.4 billion). Over the next 5 years, SPV will experience a negative cash flow each year as the preference dividend rate is higher than the dividend yield on the ordinary shares. However, the ordinary dividend is growing at a rate of 8% per year. The prime rate is 10%.

	0	1	2	3	4	5	Total
Prefdividend payable		-37.74	-37.74	-37.74	-37.74	-37.74	-188.70
Ordinary dividend rec.		22.58	24.39	26.34	28.45	30.72	132.48
Dividend withholding tax-exempt		-	-	-	-	-	-
Net Cash Flow to SPV		-15.16	-13.35	-11.40	-9.29	-7.02	-56.22

There will be no dividend withholding tax on the ordinary dividend receivable as the BEE company is exempt from DWT as it is a resident South African company. There is no DWT on the preference dividend payable for the same reason.

SPV Ltd will need to finance the annual shortfall and this may result in further finance costs and loans payable. However, if the market capitalisation is equal to R2.4 billion, then SPV Ltd's share will be equal to R1.224 billion (51% × R2.4bn). If the company pays back the preference shareholders R510m, then SPV Ltd will be left with a positive net equity value of R714m. This translates to 30% of the company (714m/2400m). However, the company may have to deduct a loan liability of R56.22m, plus interest, to cover the annual cash flow shortfall over the 5-year period. If we ignore interest, and we deduct R56.22m, then this equates to 27% of the

company and the BEE company has lost control. However, SPV Ltd should be able to retain the level of gearing and retain the control of the entity by raising further loans or undertaking a new preference share issue in 5 years' time.

If share prices fall or stay the same, then the BEE entity will be under water at the end of 5 years. Even if the market capitalisation rises to say R1.6 billion, then the net equity at the end of 5 years will be R249.78m (1 600m × 0.51 – 510m – 56.22m) and this equates to 16% of the company, which is far from 51%. However, the company should be able to raise finance to maintain control but will remain a highly leveraged entity.

What happens if interest rates rise over the period? This has two effects. Firstly, the interest payable on the loan or preference dividend rate rises resulting in higher financing costs. Secondly, the higher interest rate results in a fall in the value of ordinary equity which is the underlying asset or security of the BEE entity. This is what happened in 1998 when interest rates increased dramatically resulting in a number of BEE entities and their financial structures unwinding and the BEE equity ownership being reduced to almost nothing. The financial crisis in 2008 and early 2009 resulted in a number of BEE transactions being underwater due to the fall in equity prices. However, we have seen a substantial recovery since then due to higher share prices. However, certain sectors such as construction and cement have produced very low returns for BEE partners. In contrast, BEE investors in Vodacom and Sasol achieved good returns. The prospects for sustainable BEE financial structuring are brighter now.

Alternative BEE financing structure

In terms of Section 45 of the Income Tax Act, a company can sell its 'business' to a new company (Newco) and if the shareholders of the company hold at least 70% of the equity in Newco, then the sale will not be subject to taxes, such as capital gains tax or the tax effects of recoupments. This is a significant advantage for BEE transactions because debt can be introduced into Newco to acquire the 'business' of the company and the interest payments will be deductible as debt has been used to acquire the business of the company rather than the equity.[7] Newco will be highly leveraged but the selling company may subordinate loans due to the company.

This structure enables the BEE partner to have direct ownership in the operations of the company. Assume that Metrix Holdings Ltd which is an operating company decides to sell its business to Newco and it retains 75% ownership and

introduces a BEE partner with 25% ownership in Newco. The business is worth R1 000 million and Newco purchases the business for this price. In terms of financing, assume Metrix contributes R30m in equity financing and the BEE Company provides R10m. Newco borrows the balance in the form a long-term loan from a bank. Alternatively, Metrix may provide Newco with a shareholder's loan for the same amount. Let's assume that the interest rate on this bank loan is 10% and the business is currently generating earnings before interest and tax (EBIT) of R120m per year.

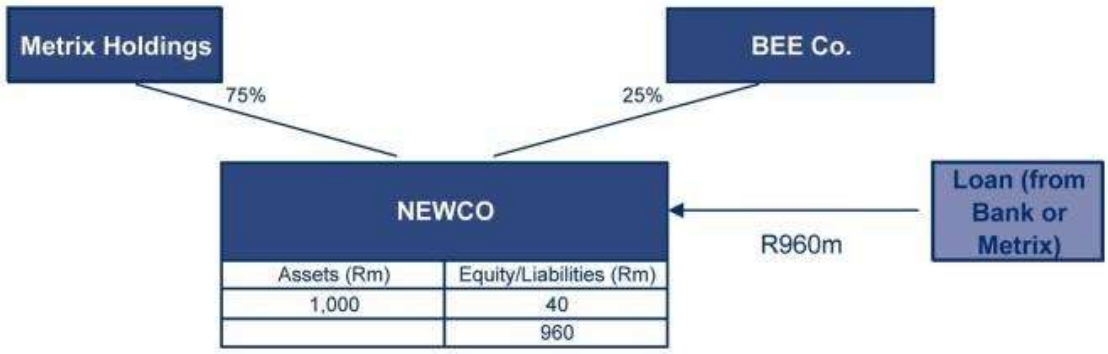


Figure 13.4 Alternative BEE financing structure

The value of the assets should grow over time and the BEE partner shares in the growth in value. However, the company is highly leveraged and Newco is required to ensure that its operating cash flows do cover the interest charge of R96m per year. If there is a downturn then Newco will be vulnerable if operating cash flows do not cover the interest charge.[8] At the end of, say 10 years, then the company should be able to refinance the loan. This is due to the growth in the value of Newco's assets, whilst the loan liability amount has remained constant. Similar to other leveraged buy-outs, Newco will not be able to finance major further expansion unless Metrix subordinates the shareholder loan due by Newco or provides some form of guarantee or security to the bank if the BEE transaction has been financed with a bank loan.

In terms of section 24O of the Income Tax Act (effective from 1 January 2013), it is no longer necessary to undertake a transfer of assets and liabilities in order to obtain an interest deduction. In terms of section 24O, if a company borrows to acquire a controlling equity interest of at least 70% in another company that is an operating company (i.e. a company that provides goods or services and operates continuously), then the interest on this loan will be deductible for the acquiring company. However, section 23N also applies to section 24O and will limit any deduction.[9] Section 23N suggests that perhaps interest of 40% of adjusted taxable income (taxable EBITDA) reflects an acceptable level of financial leverage. Highly levered structures have been

contested by SARS and in our view firms are less likely to make use of highly indebted entities to acquire assets in the future.

Financial assistance for the acquisition of shares: the requirements of the Companies Act, 2008 Vendor financing plays an important role in enabling BEE transactions and financial assistance is often crucial to any BEE transaction. In terms of section 44 of the Companies Act of 2008, a company is permitted to offer financial assistance for the purchase or subscription of shares or other securities in the company. Financial assistance also includes loans, security and guarantees. However, a company may only provide financial assistance if the following conditions are met:

- The Memorandum of Incorporation permits financial assistance;
- Authorisation by the Board of Directors has been obtained;
- Special resolution of shareholders adopted within previous 2 years, unless it relates to an employee share scheme;
- The board is satisfied that the solvency and liquidity tests are met after the assistance; and ■ The terms of the financial assistance are fair and reasonable to the company.

FROM THE REAL WORLD … SASOL AND ABSA BEE TRANSACTIONS

The **Sasol Inzalo Public empowerment scheme (SIPBEE)**, which was launched in 2008, enabled black South Africans to purchase Sasol Inzalo shares at a discounted price. Sasol Inzalo acquired preferred ordinary Sasol shares with financing provided by an external party for the majority of the purchase price over the term of scheme. On 7 September 2018, Sasol informed shareholders that Sasol had repurchased all of Sasol Inzalo's Preferred Ordinary shares at the 30-day volume weighted average price of R542.11 per share. Sasol Inzalo used the proceeds to first repay the external debt of R7.4bn, after which there was R1.377bn to distribute to the Inzalo shareholders which amounted to a cash distribution of R85.63 per Inzalo share. The scheme required the Sasol share price to be at least R462 in order to repay the debt – this was the break-even price before the Inzalo shareholders would receive any value. The fact is that the scheme was under water for many years because the Sasol share price traded below this price, but the increase in the oil price led to a significant increase in the Sasol share price from June 2018 until September 2018 and the scheme's term came to an end at an almost perfect time. The Sasol Inzalo value per share was zero for a long time and then it rose dramatically from June 2018. The price received by the Inzalo shareholders was R85.63 per share. Yet, if Sasol had acquired the Inzalo shares just two months later, the value

per share would have been about 80% lower – a dramatic loss in value. The Sasol Inzalo shares were ultimately options which involve greater volatility in value than the underlying Sasol shares. *Business Day* (12 September 2018) reported that Inzalo's public shareholders had achieved a return of 19% per annum. See Chapter 6 for more on Inzalo up to 2023.



Figure 13.5 Sasol share price relative to the final break-even price

Page 13-43 **ABSA**'s initial BEE transaction involved the issue of 73.15m redeemable option-holding preference shares with a par value of R2.00 each. ABSA issued the preference shares to Batho Bonke, which is an economic empowerment group using a company, Newco for this purpose. Newco stood to acquire 10% of the ordinary shares of ABSA. The preference shares carried full voting rights. Each preference share offered its holder the option to subscribe for one ABSA ordinary share. The exercise price was R48 if the share price is below or equal to R70. If the share price was between R70 and R100, the exercise price was R48 plus R0.70 for each R1 that the share price exceeds R70. The maximum exercise price was R69 [R48 + 0.70 × (R100-R70)], once the share price exceeds R100. The cash consideration for the preference shares of R146.3m was payable by Newco to ABSA and was redeemable within 5 years' time. The options needed to be exercised within three and five years after the transaction date. The preference shares would be redeemed at R2.00 each if the options lapse or if the options were not exercised. A dividend rate of 72% of the prime rate was payable by ABSA on the preference shares. We will explain options in greater detail in Chapter 18.

On 1 June 2009, ABSA announced that it had bought back 49.9% of the options (36.5m preference shares and associated options) at a price of R29.10 per share and had issued 36.65m ordinary shares arising from the exercise of the options attached to the redeemable shares held by Batho Bonke. ABSA announced that the proceeds were used as follows:

Application of proceeds from repurchase	Rm
Gross amount	1,062.2

Application of proceeds from repurchase	Rm
Exercise of part of the remaining 50.1% (36.65m)	-817.6
Settlement of funding and accrued dividends due to Sanlam	-74.1
Transaction and administration costs	-30.0
Payable to Batho Bonke shareholders	140.5

Batho Bonke exercised the remaining 36.65m options at R68.30 per option and ABSA issued the same number of ordinary shares in ABSA. The funding cost of R2 503.2m (36.65m × R68.30) was financed with R817.6m from the repurchase and a back up facility of R1 685.6m from ABSA. This facility was replaced with permanent financing from Sanlam, the PIC and the Development Bank of Southern Africa. In the end the transaction resulted in BEE ownership of 5.1% of ABSA. It was unfortunate that Batho Bonke did not obtain financing for all the options, rather than enter into the repurchase as this would have resulted in an increase in value of about R1 420m by 2010. However, the financial crisis in early 2009 had a negative impact on all refinancing transactions.

10 ISLAMIC FINANCE

South Africa became the first African country to issue a sovereign Shariah-compliant bond when it raised $500m in 2014, which was more than four times subscribed, and was priced at 3.9%. The bond matured in 2020 and according to the Financial Times represented a Sukuk *Al-Ijara* structure with cash flows based on infrastructure assets. South Africa's participation in sukuk issues reflects the growing importance of Islamic finance within the global financial system.

Islamic finance represents financing that complies with the principles of Islamic law, known as Shariah. It enables Muslims to undertake business and financing activities that are consistent with Shariah principles. There are three core principles of Islamic finance which relate to the role of money, asset-based financing and the sharing of risks.

Money is only a medium of exchange and value is based on trade and production. Money is not a commodity and has no intrinsic value. In conventional banking money creates money and interest or usury is the price of money. This is against Islamic law and money is only a medium of exchange and a unit of measurement.

Asset-based financing is a key component of Islamic finance. There needs to be an underlying asset for which financing is required. In conventional finance, loan transactions are often independent of assets and are interest-bearing. In Islamic finance, transactions involve the sale, transfer or leasing of an asset.

Risk-sharing is important in order to achieve a fair balance between parties to a transaction. Parties to any transaction are required to act in a fair, ethical and just manner. Aligned to this principle, parties are required to share risk in a way that is fair and are required to share in the profits and losses of a project or venture. In order to earn a return, it is important to participate in a meaningful way in the risks of a venture.

Interest (*Riba*)

In conventional banking, financing is not directly linked to an asset or project. If the project fails, the borrower is still required to repay principal and interest for the use of money. In Islamic finance, the financier is required to participate in the risk of an asset or project and any return is earned from taking on business risks and sharing in the profit or loss of a project. Islamic banking structures are required to be free of interest.

Uncertainty (*Gharar*) and Speculation (*Maisir*)

If there is excessive risk and uncertainty in respect to the terms of a contract, then such contracts are not permitted in terms of Shariah. The value of an option involves a great deal of uncertainty in respect to its value on expiry. A futures contract also involves uncertainty and therefore most derivative instruments would not be permitted in terms of Shariah. Insurance has highly uncertain outcomes and Islamic finance have adopted insurance-type structures that comply with Shariah. Gambling, lotteries and betting is prohibited by Shariah and financial speculation such as trading in currencies and trading in derivatives is not permitted.

Unethical investments (*Haram*)

Business activities and investments relating to alcohol, pornography, gambling, pork products, tobacco and armaments are not permitted as these represent unethical investments.

What are some of the financing structures and methods that comply with Shariah?

- *Murabaha* is an arrangement whereby the financier will purchase the asset and resells the asset to the client at an increased price and payment is deferred to a future date.

- *Ijara* is an arrangement whereby the financier purchases the asset and leases this to the client for an agreed rental.
- *Musharaka* represents a partnership structure whereby parties together provide labour and capital and share in the risks and profits and losses of a project.
- *Mudaraba* is a partnership whereby one partner provides capital and the other partner provides labour. In this case, only the capital partner will share in losses.

Page 13-45 Sukuk (Islamic bonds) are notes or trust certificates that represent a proportional ownership of an asset or a pool of assets. In order to comply with Shariah, there needs to be a direct ownership of assets. Income to the bonds derive from the cash flows of identifiable assets. An Islamic bond represents a real ownership stake in an asset and so the Sukuk issue by South Africa represents ownership of infrastructure assets. Equity investments are fine as these investments involve risk-sharing and participating in the profits and losses of investments. Therefore, investing in shares on the JSE is permitted but it is important not to invest in companies involved in activities such as gambling, alcohol, speculation, non-halal products and conventional banking.

FROM THE REAL WORLD ... ISLAMIC FINANCE AND THE JSE

The JSE in 2008 launched a Shariah compliant FTSE/JSE Top40 index which is impartial, fatwa- certified and subject to asset-based debt screening. The JSE states the following in its fact sheet:

> The FTSE/JSE Shariah Top 40 index is designed to reflect the performance of the top Shariah compliant companies, as screened by Yasaar, from the FTSE/JSE Top 40 Index. The Shariah is the Arabic term for Islamic Canon Law – a system of rules that is derived from the Quran and the Sunnah. According to the Islamic faith, money is not considered an end in itself but a means to higher values.

The JSE undertakes qualitative and quantitative screening to ensure that the Index is Shariah-compliant. Yasaar Limited is responsible for reviewing the Shariah compliance of the constituents of JSE indices. The JSE presents the following in the factsheet for the Shariah Top 40 Index:

Business Activity Screening

Initially, companies involved in any of the following activities will be filtered out as non Shariah- compliant:

- Conventional Finance (non-Islamic Banking, Finance and Insurance, etc.)

- Alcohol
- Pork related products and non-halal food production, packaging and processing or any other activity related to pork and non-halal food
- Entertainment (Casinos, Gambling and Pornography)
- Tobacco
- Weapons, arms and defense manufacturing

Financial Ratios Screening

The remaining companies are then further screened on a financial basis. The following financial ratios must be met for companies to be considered Shariah-compliant:

- Debt is less than 33.333% of total assets
- Cash and interest-bearing items are less than 33% of total assets
- Accounts receivable and cash are less than 50% of total assets
- Total interest and non-compliant activities income should not exceed 5% of total revenue

The Satrix Shari'ah 40 ETF (NFSH40) tracks the JSE Shariah Top 40 index. The purification of dividends through the donation to charity needs to be effected by each investor. Which companies are included in the Shar'iah ETF? Table 13.3 sets out the constituents of the Shariah compliant ETF. There are no banks and there is a significant weighting towards mining companies. Page 13-46

Table 13.3 Constituents of the Satrix (NewFunds) Shari'ah 40 ETF (NFSH40)

CODE COMPANY	WEIGHTING %
AGL ANGLO AMERICAN PLC	27.53%
AMS ANGLOPLAT	1.66%
ANG ANGLO GOLD ASHANTI	5.35%
APN ASPEN	2.74%
BHG BHP	4.48%
BVT BIDVEST	3.99%
CLS CLICKS LTD	2.74%
EXX EXXARO RESOURCES	1.80%
GFI GOLDFIELDS LTD	7.63%
GLN GLENCORE PLC	3.77%
HAR HARMONY GOLD	1.61%
IMP IMPLATS	3.39%
MNP MONDI PLC	6.74%
MRP MR PRICE GROUP LTD	1.46%

CODE COMPANY	WEIGHTING %
MTN MTN GROUP	8.74%
NRP NEPI ROCKCASTLE PLC	2.18%
SOL SASOL	6.44%
SSW SIBANYE STILLWATER	3.47%
VOD V0DAC0M	2.72%
CASH	1.57%
	100.00%
Industry	**Weighting %**
Basic Materials	58.88%
Telecommunications	11.46%
Industrials	10.73%
Chemicals	6.44%
Health Care	2.74%
Personal Care, Dmg & Groceries	2.74%
Energy	1.80%
Retail	1.46%
Real Estate	2.18%
Cash	1.57%
Total	100.00%

Shari'ah Top 40 ETF Basket Constituents Weightings as of 3 October 2023

GUIDANCE WHEN PRESENTED WITH PROBLEMS RELATING TO SOURCES OF FINANCE

By Johnathan Dillon M.Com CA(SA)

A person can gain a sound knowledge of the sources of finance content covered within this chapter. However, when presented with a practical problem relating thereto, knowledge of content is important but unlikely to be sufficient in order to adequately solve such a problem. Understanding how the sources of finance content link together and understanding the crux of the problem at hand are essential components required to solve practical sources of finance problems.

Understanding how the sources of finance content links together

As a basis for understanding how the sources of finance content covered in this chapter link together (especially with content covered in other chapters), a summary of key issues that need to be considered when presented with sources of finance problems is presented below:

- *The cost of sources of finance:* A company would naturally prefer a cheaper source of finance. This aspect usually requires calculating and comparing the effective cost of the various sources of finance (IRR). However, cheaper is not always better as the other factors listed below, amongst others, need to also be considered.

- *Maturity and cash flow matching:* Consider what the finance raised is going to be used for. If it is required to purchase a capital asset or to fund a permanent level of current assets then long-term finance, such as equity or a long-term loan, is preferred. However, if a short-term project or fluctuating current assets need to be financed then a short-term source of finance, such as a bank overdraft, will most likely be applicable. Furthermore, it is important to match the cash flows generated over the useful life of the asset financed with the repayments required to be made on the source of finance. For instance, if most of the cash inflows from a capital project occur in later years, then a source of finance that does not require substantial repayments in the initial years is preferable. Maturity matching ensures that the cash inflows from the asset being financed match the cash Page 13-47 outflows of the source of finance as best possible. Further considerations linked to this point include the following:
  - Local versus foreign finance: A foreign source of finance is usually only considered a worthwhile option if cash inflows generated are denominated in the same foreign currency - this will lead to a natural hedge and eliminates the need to hedge and the associated costs thereof. If this is not the case, a foreign source of finance may nonetheless be a viable option if it is cheaper after taking into account the cost of hedging the associated foreign exchange risk.
  - Variable versus fixed interest rates: This is often an important consideration when comparing various sources of debt finance. Variable interest rate loans carry more risk as interest payments change when market interest rates change but they are nonetheless used if it is predicted that interest rates will come down or remain at a relatively low level. Variable rate loans are also usually cheaper than fixed interest rate loans. However, fixed rate loans are normally chosen due to the certainty regarding cash outflows. If a company earns interest income at a variable rate then there is the opportunity to raise debt finance with interest payable at a variable rate and naturally hedge the company's interest rate risk exposure (that is, matching the basis of interest income with interest expense).
- *The level of business risk:* This relates to the certainty of cash flows expected to be generated from the asset being financed. If future cash inflows are uncertain then equity finance is safer as there is no requirement to pay a return thereon; however, if the cash inflows are fairly certain then debt finance is a better option as it is cheaper. It is important that the operational (business) risk of a project is considered when deciding on the source of finance. This is done in order to balance the expected or current level of business risk with any possible increase in financial risk arising from a particular source

of finance and it ensures that the company's total risk exposure (considering business and financial risk) is not excessive.

- *Covenants, restrictions and/or regulatory requirements:* Loan covenants or other legal restrictions may be in place, restricting a company's ability to raise further debt finance. In such an instance raising additional equity finance may be the only option. Alternatively the company could obtain the use of an asset by means of an off-statement of financial position operating lease (as this will not alter the company's book capital structure). However, this may not be possible as certain loan agreements prohibit the borrower from entering into any lease agreements without the prior consent of the lender.
- *Dilution of control and earnings per share:* Based on the current shareholdings and preferences of shareholders, this may be an important consideration for current shareholders in which case debt finance would be preferred. Earnings per share can also be levered higher through the use of debt finance provided the return on capital employed is in excess of the cost of debt.
- *Security:* If the company has material assets which are not security for any other loans then there is an incentive to put these assets up as security so as to reduce the cost of debt finance. This needs to be weighed up against the disadvantages associated with the company's material operating assets being secured.
- *Capital structure considerations and the company's current and target level of financial risk:* The company's current level of financial risk (capital structure) needs to be considered. For instance, if the current capital structure has too much debt (excessive financial risk, generally evidenced by a debt ratio in excess of its target capital structure) then equity finance will be a better option if further funding is required. For further information on this point, refer to the guidance on capital structure provided in Chapter 14.
- *Signalling effects:* Cognisance needs to be taken of the signal sent to the market which occurs when a particular source of finance is raised – if equity finance is raised then generally the signal sent to the market is that the company has 'poor' or uncertain future prospects, while the raising of debt finance generally sends the signal that the company has 'good' or certain future prospects. The latter signal is based on the premise that a company only raises further debt finance if it is certain that it can cope with the additional financial risk and earn a future return in excess of the cost of debt.

The above list is not exhaustive. It does, however, raise important issues that need to be considered when presented with sources of finance problems. A common example thereof is the need to recommend the most appropriate source of finance given a particular scenario. If such a situation requires the comparison of two or more sources of finance then it is

important to take careful note of the various terms of the given sources of finance and to compare them considering the guidance provided above. Furthermore, it is important to understand that the above pointers are not mutually exclusive and could all apply within a given scenario. A lack of understanding of the above issues is likely to result in someone not identifying all the pertinent issues that need to be considered before making a final recommendation.

Understanding the crux of the problem

Often a lack of understanding with regard to what is required in a problem relating to sources of finance results in someone either missing the point or not knowing what to do. Ask yourself "what is the root of the problem at hand that I need to solve?" The problem will generally be one of the following:

1. *A discussion on which sources of finance are available and could/should be used* – this includes analysing the company requiring finance and its particular circumstances, and then recommending possible sources of finance (for example, equity share capital, preference share capital, long-term loans, debentures, bonds, bank overdraft, factoring and so on).
2. *An analysis of potential sources of finance when presented with available financing options* – this includes analysing the company requiring finance and its particular circumstances, and then recommending which available source of finance it should select (for example, a comparison of preference shares and debentures which could be issued).
3. *A discussion of where to obtain finance, usually given a specific source of finance e.g. where to obtain long-term loan finance* – this includes analysing the company requiring finance and its particular circumstances, and then suggesting from which institution finance could/should be obtained (for instance, from either a commercial bank, the Industrial Development Corporation, Business Partners and so on).

It is important to note that each of the above points require an analysis of the company and its particular circumstances – this is crucial as any practical problem cannot be solved without due regard for the set of facts at hand. Furthermore, and in conclusion, once the crux of the problem has been identified it is important to solve the problem considering the aforementioned key issues.

SUMMARY

A business may be thought of as comprising two distinct aspects, operations and financing. A business needs finance to support its operations. Finance is either raised in the money market or the capital market.

There are various institutions that comprise these markets, ranging from the different types of banking institutions through to the JSE Equity Market, JSE Debt

Market (BESA) and the informal markets. The main sources of finance are equity and debt. Equity means ownership. Debt contracts will refer to the term, indicate a fixed or variable interest rate and will include loan covenants.

Although debt is less costly than equity and may result in the tax deductibility of interest, it is also riskier. Issuing debt has less impact on the control of a company than equity. Page 13-49

Inflation linked bonds offer real returns and will protect the returns of investors from the effects of inflation. Technology has enabled the launch of online marketplaces for lending and borrowing as well as providing a marketplace to invest in start-up firms and projects. BEE financing structures may involve vendor financing, loans and/or preference shares.

SELF-STUDY PROBLEMS

S13.1

Two firms, A and B, need to raise finance. A is in a high-growth business, with good prospects for future projects. The business is volatile and cash flows are difficult to predict. B operates in a stable mature industry.

1. All else being equal, which firm should borrow more?
2. Consider A only. The owners hold 55% of the issued share capital. How does this affect the decision to raise funds through equity?

S13.2

Mogale Ltd, a hardware supply company, is considering expanding its operations to KwaZulu- Natal. The company wishes to make a rights issue to raise the required funding of R24 million. The company is listed on the JSE and has currently 9 million shares in issue. The current share price on the JSE is R10 and the subscription price will be set at R8 per share.

Required:

(a) How many shares will be issued in terms of the rights issue?

(b) What is the expected ex-rights price?

(c) What is the value of each right?

Solutions to self-study problems

S13.1 Solution

1. B can afford to borrow more as its cash flows and needs are more predictable, hence there is greater certainty about its ability to service debt. A could face difficulties if the volatility leaves it in a position where it cannot meet its interest or capital repayment obligations.
2. The owners would probably be reluctant to lose control. However, loss of control is less damaging than loss of the firm through insolvency. The owners could consider a hybrid which might protect them against the volatility of the industry while at the same time would result in fewer contractual obligations. Examples include convertible debentures and preference shares. However, these would have to be priced to make them attractive to potential investors.

S13.2 Solution

1.

Number of shares to be issued	
Funding from rights issue	24,000,000
Subscription price	8.00
Number of shares to be issued	3,000,000

Page 13-50 **2.**

Number of shares to 1 right			
Number of shares to be issued			3,000,000
Total number of shares prior to rights issue			9,000,000
Number of shares to 1 right			3.0

Ex-rights price	Share price	No. of shares per right	
Value of shares prior to rights issue	10.00	3	30.00
Subscription price			8.00
			38.00
Number of shares held after the rights issue			4.00
Estimated ex-rights price			**9.50**

Alternatively, we can determine the ex-rights price as follows:

	Share price	No. of shares	Total Value
Value of shares prior to rights issue	10.00	9 000 000	90 000 000
Financing from rights issue			24 000 000
Total value			114 000 000

	Prior to issue	Rights issue	
Total number of shares after the rights issue	9 000 000	3 000 000	12 000 000
Estimated ex-rights price (114m/12m)			9.50

3.

Estimated ex-rights price	9.50
Subscription price	-8.00
Value of Right	**R1.50**

QUESTIONS

Question 13.1

Explain the reason for the great variety of debt and equity instruments that are available.

Question 13.2

What are the advantages to be obtained from a listing on the stock exchange:

(a) for the company;

(b) for the major shareholder?

Question 13.3

‘Preference shares are a hybrid of debt and equity and contain the best features of each as far as the issuing company is concerned. It follows that companies should make extensive use of preference shares for their long-term financing requirements.’ Discuss. Page 13-51

Question 13.4

What are some of the issues involved in determining whether a firm should select a variable interest rate loan or a fixed interest rate loan?

Question 13.5

Shareholders and the suppliers of loan finance both have a right to the operating income of the company. In which ways do these rights differ? Which is preferable from the company’s point of view?

Question 13.6

How does higher financial leverage affect the following?

(a) Earnings per share

(b) The earnings stream

(c) Operating leverage

Question 13.7

Explain what you understand by primary and secondary issues in the capital markets.

Question 13.8

Moqhaka Ltd is considering the investment in new plant and equipment at a cost of R18m. The company will pay a deposit of R3m. The bank has offered the firm a loan of R15m, repayable in equal annual instalments over 5 years at an interest rate of 9%. The economic life of the project is 5 years and the company expects to achieve a residual value of R6m on the sale of the equipment at the end of 5 years. Alternatively, the bank is also prepared to offer a loan of R15m, repayable in equal annual instalments except for a bullet or balloon payment at the end of 5 years, equal to R6m, which the company will pay from the proceeds on sale of the equipment. The interest rate is also 9% on this loan. What is the annual instalment required under either loan alternative? What is the interest and capital repayment per year under either alternative?

Question 13.9

Refer to the rights issue example in the chapter. Illustrate how the net wealth of a shareholder who owns 900 shares remains the same regardless of whether she decides to sell the rights or accept the offer.

Question 13.10

Pharmagen Ltd is undertaking major research and development and is making a rights issue to raise the required financing. The company has currently 30 million shares in issue. The company needs to raise R42 million. The share price has been trading within a range of R11.00 - R15.00 in the last year, but the latest quoted price is R12.00 per share. The subscription price for the rights issue will be R9.00 per share.

Required:

(a) How many shares will the company issue?

(b) How many shares will a shareholder need to hold to acquire one right?

(c) What is the expected ex-rights price?

(d) What is the value of a right?

Question 13.11

Curtin Holdings Ltd wishes to raise R100 million through a rights issue. The funds are required to fund newly developed groundbreaking medical technology. Curtin Holdings Ltd has 100 million shares in issue which currently trade at R2.50 per share. Fred Curtin, the company secretary, has suggested a subscription price for the shares of R2.20 under PLAN X or R1.80 under PLAN Y. You are unsure which plan to choose. Page 13-52

Required:

(a) How many shares is a shareholder required to hold to purchase a new share under either plan?

(b) Calculate how many shares are issued under both plans.

(c) Under each plan, what is the ex-rights price?

(d) What is the value of a right under PLAN X and PLAN Y?

(e) From the company's point of view which plan is preferable?

(f) Which plan would you recommend?

(g) Would it be smart to ask shareholders for a subscription price of R3 per share?

Question 13.12

Northstar Ltd requires raising R200 million to develop a new lithium mine in the Northern Cape. The company is listed on the JSE and is currently trading at a price of R15.00 per share. The company wishes to raise the required finance by making a rights issue to existing shareholders. The company's statement of financial position reflects a shareholders' equity of R500 million and the company currently has 400 million shares in issue.

Required:

(a) Explain why the maximum subscription price is R15.00 per share.

(b) What could be an issue with setting a subscription price very close to the current listed price?

(c) Assume the company sets a subscription price of R12.00 per share. How many shares will the company issue?

(d) What is the ex-rights price?

(e) What is the value of a right?

(f) Why is it important that a shareholder sell his rights if he is unable to take up his rights?

Question 13.13

Ndebele Mining Ltd has recently made a rights issue of 1 million shares at a subscription price of R9.00 per share. The objective was to raise funds to finance a new mining development. The ex-rights share price is R10.00. There were 6 million shares outstanding prior to the rights issue. What was the share price of Ndebele Mining prior to the rights issue?

Question 13.14

Isiduko (Pty) Ltd has 1 million shares in issue, 70% of which are held by Ann Isiduko. Her cousin, Mandisa, holds 20% and three individuals important to the operations of the company hold the other 10%.

Isiduko needs to raise capital of R30 million. An investment banker has indicated that the money could be raised by the placement of shares at R31.50 each to net Isiduko R30.00 each. The investment bank will earn R1.50 per share. What factors should Ann Isiduko consider in evaluating the issue of additional shares? Support your comments with calculations where appropriate.

Question 13.15

Mr A has invented a radically new item of sports equipment and has financed the development and prototype stages from his own resources. Market research indicates the possibility of a large volume of demand. A significant amount of additional capital will be needed to finance production.

Required:

Advise Mr A on:

(a) the advantages and disadvantages of loan or equity capital from his own point of view;

(b) the various types of capital likely to be available under the circumstances of this case and the source(s) from which they may be obtained; and

(c) the method(s) of finance likely to be most satisfactory to both Mr A and the supplier of funds.

(*CIMA - adapted*) Page 13-53

Question 13.16

List the advantages and disadvantages of obtaining a listing on the JSE Securities Exchange.

Question 13.17

The Statement of Financial Position of Pickwise Stores Ltd, a supermarket chain, is summarised below:

	Rm
Non-current assets	1,073
Current assets	1,817
	2,890
Equity	626
Non-current loans	215
Current liabilities	2,049
	2,890

The issued share capital consists of 156 million ordinary shares currently trading at R8.00 each, and 35 million preference shares trading at R7.30 each. Included in current liabilities is R1649m in trade payables and R400m in short-term borrowings. Current assets consist of inventory and a small balance of trade receivables. A consultant has criticised the company's financing, noting that the debt ratio is high relative to the average across all companies.

Required:

Comment on the consultant's assessment. Provide supporting calculations.

Question 13.18

You are the financial director of a company, Motheo Ltd. The following has been extracted from the company's Statement of Financial Position:

Statement of Financial Position as at 31 December (Rm)

Equity	80	Non-current assets	160
Debt	120	Net working capital	40
	200		200

Number of shares in Issue	10m
Current price per share	R4.00
Total operating income (EBIT) for the year ended 31 December	R24m

The firm is considering a project costing R22m with a net present value of R8m. The project is expected to increase earnings before interest and tax (EBIT) in the first two years by R600 000 per year but EBIT is expected to further increase thereafter. The company is perceived to be overgeared, and you fear that further use of debt will lead to a downgrade in the firm's credit rating. You believe that this project should be financed by a new equity issue, but unfortunately the market has been depressed for a while. The interest rate is 15%. The corporate tax rate is 27%.

The managing director argues that the project should be delayed for three reasons:

(a) It is too expensive for the firm to issue new debt.

(b) Financing the project with new equity will reduce earnings per share because the market value of equity is less than the book value.

(c) Equity markets are currently depressed. If the firm waits until the market index improves, the market value of equity will exceed the book value and equity financing will no longer reduce earnings per share.

Required:

Evaluate the managing director's reasons for delaying the project. Provide supporting calculations of the effects on earnings per share of adoption of the project and issue of new shares.

Question 13.19

Thelma Durandt is about to start an earth-levelling operation for which she requires capital of about R100m to buy the equipment and operate the company. Earnings before tax and interest (EBIT) are expected to amount to R20m (the tax rate is 27%). The funds can be raised by issuing shares for the full amount or, alternatively, half the requirements may be raised through the issue of 10% debentures.

Required:

Calculate the return on equity under both alternatives (ignore issue expenses). What factors should be considered in deciding on which financing alternative to adopt?

Question 13.20

Frito Misto Pty Ltd operates five trawlers off the West coast. Salvatore Pescatore started operations with a single boat and continued in this manner for many years, earning the Pescatore family an adequate living.

Ten years ago, Salvatore's son, Alfio, was awarded the MBA degree and his daughter, Santuzza, qualified as a chartered accountant. Both joined their father in his business. Through their efforts they showed how the business could grow by borrowing funds to finance further trawlers. Salvatore would not hear of borrowing money from a bank, but he was persuaded at least to borrow the money from the family. Loans were raised from cousins Francesco and Leonardo who had made a lot of money in the security business, and even Salvatore's sister Zia Pina and his mother Nonna Gerba were able to provide substantial loans from monies they received from Sicily.

A company was formed and one million shares were issued: 55% went to Salvatore, and 20% went to each of the children. Mamma Lucia, Salvatore's wife, received 5%. Since the formation of the company and the introduction of the financial and general managerial skills of Alfio and Santuzza the company has prospered and profits have now reached R4 million per annum before tax.

A ship has become available at a cost of R10m and the children feel that it should be acquired. However, the company, the shareholders, and the family, will be able to supply only R5 million. Despite the fact that Salvatore has mellowed as he has aged and is not as locked into the manners of his heritage as he used to be, he still holds the view that to borrow from a banker is dishonourable.

A number of gentlemen, always dressed in dark pin-striped suits, have been seen asking questions around the boats and talking to the staff. They are representatives of another family who operate a financing business known as the Banco di Castellamare del Golfo in Cape Town.

It is rumoured that they are prepared to acquire a substantial stake in the business at a premium (the shares in Frito Misto are worth about R5 each). It is further believed that while the fishing operation will continue, the Banco is particularly keen to have its own ships for diversifying into the import/export business for specialised products.

Required:

Identify all the courses of action open to the Pescatore family. Explain why you believe each may or may not be a good idea and whether it is likely to succeed.

Question 13.21

The main points of negotiation between the company and the investment banker when issuing debt instruments are the interest rate, the term, the issue price, and the redemption price. On which of these points would you place most emphasis during your negotiations:

(i) during a recession

(ii) during a boom period

(iii)if your company is approaching maturity

(iv)if your company is in a growth phase

(v) if your company generates large periodic cash inflows?

Question 13.22

AfriViews Pty (Ltd) ('AfriViews') is a family-owned company specialising in the manufacturing and installation of aluminium and glass. The company had experienced significant revenue growth over the three years prior to the 20x0 financial year (FY20x0). To keep up with customer demand, AfriViews has aggressively expanded the business by purchasing more assets and increasing the number of employees from 46 to 153. The capital expansion was financed by utilising debt, and AfriViews' debt to equity ratio was 22% at the end of FY20x0. AfriViews is required to repay a R5m loan from the CFO at the end of 20x5. Interest of 6.5% is charged, but is not payable annually on this loan, but will be added to the outstanding loan capital, compounded annually and will be settled together with the capital amount on 30 June 20x5. The CFO owns 49% of the ordinary equity of the firm whilst a private equity firm, CAI, owns 51%. The financial year-end is 31 December. The CEO, Mr Patrick Sabelo who is the majority owner of CAI, proposed the following two additional financing options to assist AfriViews achieve its strategic objectives and maintain its long-term viability.

Option 1: Long-term loan

XYZ Ltd, a registered financial service provider specialising in finance for small and medium enterprises, was prepared to grant AfriViews a R20 million loan on 1 January 20x1. The loan is repayable on demand if AfriViews does not comply with principle 7 of King IV. The loan would bear interest at the prime lending rate plus 2%, payable on

31 December each year. The current date is 31 December 20x0. The current prime rate is 7.25%. Interest would be compounded annually. The loan would be payable at the end of five years. Forecast prime lending rates are as follows:

20x1	20x2	20x3	20x4	20x5
6.75%	7.25%	7.50%	7.75%	7.75%

Option 2: Preference shares

CAI was willing to subscribe for preference shares to the value of R20 million. AfriViews would issue the shares on 1 January 20x1. The preference shares would carry no voting rights. The preference shareholders would be entitled to a fixed dividend payment payable annually in arrears and calculated at 5% of the nominal value. The preference shares would be redeemable at a premium of 15% after five years.

Required :

(a) With reference to the two financing options, calculate the most cost-effective financing option (employing internal rate of return)

(b) Recommend with reasons (including an analysis of the characteristics of each option) which of the two financing options AfriViews should select.

(*SAICA ITC 2021 – adapted. This is limited to the section on financing options*)

Question 13.23

ABC Ltd produces a single product which is a high-technology component used only by the motor industry. Owing to the cyclical nature of this industry, sales volumes fluctuate widely from year to year. One hundred thousand of these components were sold to the motor industry at R400 each during the year ended 31 March 20.6. Page 13-56

The company's Statement of Comprehensive Income for the year ended 31 March 20.6 is as follows:

		R000
Sales		40,000
Less: Variable costs	16,000	
Less: Fixed costs	8,000	24,000
Earnings before interest and tax		16,000
Less: Interest		5,000
Net income before tax		11,000
Tax	30%	3,300
Net income		7,700

Shares in issue (in thousands)	4,000
Earnings per share	1.925

As a result of difficulty in obtaining skilled labour, the directors of ABC Ltd are considering introducing a new production process, with used machinery, which will be more automated but highly capital intensive. It would result in annual fixed costs being doubled, but the variable cost per unit would be halved. The new process will not change the maximum capacity of the plant, which is 125 000 units per annum. The company's effective and marginal tax rate is 30%.

Two alternative methods of financing the net additional capital cost are available to ABC Ltd:

(i) raising a long-term loan at an annual interest cost of R2.8m; or

(ii) issuing 800 000 new shares.

Required:

(a) Calculate the profit before tax and interest, and earnings per share, in respect of:

  (i) the existing plant at an annual sales level of 105 000 units;

  (ii) the proposed new process financed by loan capital at annual sales levels of 100 000 and 105 000 units respectively; and

  (iii) the proposed new process financed by share capital at annual sales levels of 100 000 and 105 000 units respectively.

(b) Calculate the break-even unit sales for the three options in (a)(i), (ii) and (iii) above.

(c) Make reasoned recommendations to the directors of ABC Ltd based on the information given and your calculations, as to whether or not they should introduce the new process.

(d) State the relative advantages and disadvantages of the two financing methods, assuming the directors have decided to introduce the new process.

(*QE – adapted*)

Question 13.24

Boly Ltd is planning to revamp its operations over the coming years. It is estimated that funds of R30m will be required. Boly's latest Statement of Financial Position is as follows:

	Rm
Non-current assets	
Land and buildings	171
Plant and equipment	21
	192
Current assets	120

	Rm
	312
Share Capital	234
Debentures (10%)	30
Current liabilities	48
	312

Page 13-57 The Statements of Comprensive Income for the last three years are as follows:

	20.1	20.2	20.3
	Rm	Rm	Rm
Sales revenue	322	273	365
EBIT	41	37	47
Interest expense	3	3	3
Profit before tax	38	34	44
Tax	10	9	12
Net profit after tax	28	25	32
Dividend	14	14	14
Retained income for the year	14	11	18

Forty-eight million shares are in issue, of which the managing director, George Sorel, effectively controls 25 million. Sorel bought the company from its founders 10 years ago, and five years later obtained a listing on the JSE. Mr Sorel inherited a reasonable estate from his father and has increased its value substantially. His preferred method of operation has been to acquire small- to medium-sized businesses and develop these into large companies. Besides being the CEO of Boly, George is also on the board of several listed companies. George has three children, his eldest son is a professor of geophysics, his second son is an artist whose exhibitions are attracting favourable comment, and his daughter is completing her specialisation in psychiatry. Industry averages are:

Current assets:current liabilities	2.2:1.0
Debtors collection period	45 days
Total assets:turnover	1.0:1.9
Current liabilities:total assets	0.22:1.0
Long-term debt:total assets	0.15:1.0
Preference shares:total assets	0.05:1.0
Times interest earned	seven times
Earnings / turnover	3.5%
Return on assets	6.0%
Industry growth rate	6.0%

Boly's share price is around R4 per share. Debentures can be raised at 10%, and preference shares at 11%. The corporate tax rate is 27%.

Required:

Make a recommendation regarding the finance required.

Question 13.25

Toces Ltd is a retail company that has traded for five years. Summaries of the company's latest financial accounts are shown below. Page 13-58

Statement of Comprehensive Income		**20.1**	**20.0**
for year ending 30 June		Rm	Rm
Sales turnover		2,420	1,920
Cost of sales		-856	-650
Gross profit		1,564	1,270
Other expenses		-1,296	-1,041
Profit before taxation		268	229
Taxation		-67	-57
Profit after taxation		201	172
Dividends		-50	-50
Retained income for the year		151	122

Statement of Financial Position	Note	**20.1**	**20.0** Rm
as at 30 June		Rm	
Assets			
Non-current assets		2,728	2,460
Current assets			
Inventory		873	829
Debtors		12	10
Cash		6	10
		3,619	3,309
Equity and Liabilities			
Share Capital	1	691	691
Retained earnings		507	356
Shareholders' interest		1,198	1,047
Long-term loan (20.6)	2	1,250	1,250
Current liabilities			
Accounts payable		740	682
Short-term loan (bank overdraft)		386	290
Taxation		45	40
		3,619	3,309

1. The Toces family still owns about 36% of the ordinary shares. There are 1000 million shares in issue.

2. Interest payable: 20.1 R159.5m (20.0 R131m)

The company wishes to expand and hopes to raise R2 billion. R500m would be used to establish new shops, R250m for increased working capital, and Rl 250m to repay the existing floating rate term loan.

Toces' directors have contacted several institutions and have been offered a R2 billion 8.5% per year fixed-rate secured loan from an insurance company and a 50m Swiss franc loan at a fixed interest rate of 4% per year by a Swiss bank. The insurance company requires the option to purchase 500m new ordinary shares at 60 cents per share as part of the deal.

A third offer has been made by a venture capital company which would provide a Rl billion five-year loan at the fixed interest rate of 9% per year, and Rl billion in equity finance (1 000m new ordinary shares at a price of R1 per share).

All loans are repayable at the end of five years and interest is payable annually at the end of the year.

Inflation in Switzerland is expected to be at 2% per year less than in South Africa for the next few years. The current exchange rate is SF1 = R20.

Required:

(a) What factors should Toces' directors consider before deciding how to finance the proposed expansion?

(b) Evaluate Toces' plans for using the R2 billion.

(c) Discuss the advantages and disadvantages to Toces of each of the three financing offers. Relevant calculations must be shown. State clearly any assumptions that you make.

(d) If none of these three offers were accepted by Toces, suggest alternative sources of finance that might be available to the company.

(*ACCA – adapted*) Page 13-59

Question 13.26

THP Co is planning to buy CRX Co, a company in the same business sector, and is considering paying cash for the shares of the company. The cash would be raised by THP Co through a 1 for 3 rights issue at a 20% discount to its current share price. The purchase price of the 1 million issued shares of CRX Co would be equal to the rights issue funds raised, less issue costs of R3.2m. Earnings per share of CRX Co at the time of acquisition would be R4.48 per share. As a result of acquiring CRX Co, THP Co expects to gain annual after-tax savings of R960 000. THP Co maintains a payout ratio of 50% and earnings per share are currently R6.40 per share. Dividend growth of

5% per year is expected for the foreseeable future and the company has a cost of equity of 12% per year.

Information from THP Co's Statement of Financial Position:

Equity and liabilities	R000
Share capital (3m shares in issue)	30,000
Retained earnings	43,000
	73,000
Non-current liabilities (8% loan notes)	50,000
Current liabilities	22,000
Total equity and liabilities	145,000

Required:

(a) Calculate the current ex dividend share price of THP Co and the current market capitalisation of THP Co using the dividend growth model.

(b) Assuming the rights issue takes place and ignoring the proposed use of the funds raised, calculate:

(i) the rights issue price per share;

(ii) the cash raised;

(iii) the theoretical ex rights price per share; and

(iv) the market capitalisation of THP Co.

(c) Using the price/earnings ratio method, calculate the share price and market capitalisation of CRX Co before the acquisition.

(d) Assuming a semi-strong form efficient capital market, calculate and comment on the post acquisition market capitalisation of THP Co in the following circumstances:

(i) THP Co does not announce the expected annual after-tax savings; and

(ii) the expected after-tax savings are made public.

(e) Discuss the factors that THP Co should consider, in its circumstances, in choosing between equity finance and debt finance as a source of finance from which to make a cash offer for CRX Co.

(*ACCA-adapted*)

Question 13.27

Smith (Pty) Ltd is a manufacturer of pharmaceuticals. It lost a considerable sum of money in August, 20.6, when judgment was given against the company in a public liability suit. At that time the company was inadequately insured against the risks arising out of the side effects of the drugs it manufactures. Adequate cover has subsequently been arranged.

This loss, together with the unusually high development costs of a new drug, which is very effective in the treatment of a rare ailment, has put Smith (Pty) Ltd in extreme financial difficulties. The abridged Statement of Financial Position of the company at 31 December, 20.6, together with comparative figures, is as follows: Page 13-60

Statement of Financial Position	20.6	20.5
ASSETS	**R000**	**R000**
Non-Current Assets		
Property (at cost, subject to a mortgage bond)	9,000	9,000
Plant & equipment	7,000	7,500
Patents	3,000	2,500
	19,000	19,000
Investments	3,000	3,000
Current Assets		
Inventories	8,000	10,000
Receivables	5,000	9,000
Total Current Assets	13,000	19,000
Total Assets	**35,000**	**41,000**
EQUITY AND LIABILITIES		
Equity		
Share Capital (15m shares in issue)	15,000	15,000
Accumulated Loss/Retained earnings	-17,000	4,000
	-2,000	19,000
Redeemable preference shares - 12%, cumulative	10,000	10,000
Non-current Liabilities		
Long-term borrowings - 10% mortgate bond	5,000	5,000
Current Liabilities		
Trade and other payables	19,000	7,000
Short-term borrowings	3,000	0
Total current liabilities	22,000	7,000
Total Equity and Liabilities	**35,000**	**41,000**

The current value of the property is R16m and the investment in the associated company is valued by the directors at R6m. The other assets are fairly valued. The Statement of Comprehensive Income for the year ended 31 December, 20.6, together with comparative figures, is as follows:

Statement of Comprehensive Income	20.6	20.5
for the year ended 31 December	**R000**	**R000**
Sales	50,000	45,000
Net income before development costs and public liability claim	33,000	36,000
Public liability claim	-27,000	-
Development costs	-25,000	-8,000
Net income before tax	-19,000	28,000

Statement of Comprehensive Income	20.6	20.5
Taxation	-	-7,500
Net income after tax	**-**19,000	20,500
Retained income at beginning of year	4,000	1,200
Dividends	-2,000	-17,700
Preference	-600	-1,200
Ordinary	-1,400	-16,500
Accumulated loss/retained income at end of year	**-17,000**	**4,000**

(i) Interim dividends were paid in June 20.6, at which time the board was confident that it would win the pending action. Page 13-61

(ii) Future development costs are expected to amount to R8m per annum. The company's policy has always been to write off these expenses when incurred.

(iii) A small number of creditors are pressing for payment and are threatening to institute liquidation proceedings.

(iv) An investment group has expressed an interest in taking up 10 million new ordinary shares at R8 per share. A condition is that at least 50% of distributable income should be distributed as a dividend in future. The controlling shareholder is very enthusiastic as this transaction will save the company and leave him with 54% of the equity.

(v) The controlling shareholder has approached you for advice. He holds 90% of the shares and has only some R1m ready cash with which to take up any additional shares.

(vi) Assume that sustainable net income (before development costs) is R38m per year. The average earnings yield for listed pharmaceutical companies is 7.5%. The company is highly dependent on the controlling shareholder.

Required:

Set out what further enquiries you would make and what advice you would give to the controlling shareholder regarding his best course of action by considering the approach of the investment group and comparing it with other financing opportunities available.

(*SAICA ITC/QE – adapted*)

Question 13.28

You have been approached by the management of Jet Pack (Pty) Ltd ("Jet Pack"), a company engaged in the manufacture and sale of metal containers, for assistance with certain matters regarding a proposed management buy-out of the business. Management has no meaningful financial resources.

Parnel International, a US corporation, which owns 64.7% of the shares in Jet Pack, is engaged in a process of rationalisation. As part of this rationalisation it has decided

to dispose of some of its foreign investments. The investment of Parnel International in Jet Pack is one of those being sold. The remaining 35.3% of Jet Pack's shares are owned by Park Ltd, a listed South African company. While Park Ltd has a pre-emptive right in respect of the shares held by Parnel International, its board has indicated that it does not intend to exercise this right and that it would be prepared to sell its holding in Jet Pack if the offer by management is sufficiently attractive. It would also be prepared, if the offer is not attractive, to remain as a 35.3% shareholder.

The actual and forecast consolidated trading results of Jet Pack for the year ended 31 December are as follows (R000s):

	Audited				Budgeted	Forecast	
	20.4	20.5	20.6	20.7	20.8	20.9	20.10
Sales	290,900	318,130	341,520	430,110	473,760	541,070	616,840
After-tax profits	35,186	33,968	37,577	43,022	49,784	57,498	65,940

The 20.8 to 20.10 budgeted and forecast results do not take account of the proposed buyout. Jet Pack has one subsidiary in which there is a 15% minority interest. The only asset of the subsidiary is a property which is occupied by Jet Pack. The subsidiary's issued share capital is R2m and the balance of the funding has been provided equally by the minority shareholders and Jet Pack on loan account. The abridged consolidated Statement of Financial Position and Statement of Comprehensive Income of Jet Pack and its subsidiary for the year ended 31 December 20.7 were as follows: Page 13-62

Statement of Financial Position	20.7	20.6
at 31 December, 20.7	**R000**	**R000**
ASSETS		
Non-Current Assets		
Property, plant & equipment	79,330	72,470
Investments	2,510	3,360
Current Assets	124,300	113,040
Total Assets	**206,140**	**188,870**
EQUITY AND LIABILITIES		
Equity		
Share Capital	29,630	29,630
Retained earnings	118,550	103,800
	148,180	133,430
Outside shareholders' interest	3,000	3,000
Non-current Liabilities		
Long-term borrowings	7,060	7,060
Deferred tax	3,050	3,140

Statement of Financial Position	20.7	20.6
	10,110	10,200
Current liabilities	44,850	42,240
Total Equity and Liabilities	**206,140**	**188,870**

Statement of Comprehensive Income	20.7	20.6
for the year ended 31 December	**R000**	**R000**
Sales	430,110	341,520
Eamings before interest and tax	62,370	53,210
Interest	-910	-1,020
Net income before tax	61,460	52,190
Taxation	-18,438	-14,613
Net income after tax	43,022	37,577
Dividend	-28,272	-22,250
Retained income for the year	**14,750**	**15,327**
Cash flow from normal trading operations	51,626	41,334

The following information is pertinent to the proposed buy-out:

(i) The consolidated Statement of Financial Position includes the subsidiary's land and buildings at book value of R34.12m. At 31 December 20.7 the group had no borrowings except from the minority shareholder in the subsidiary, and it currently holds R5m in surplus cash.

(ii) The intended purchase consideration for Parnel International's share is 7.5 times the earnings of 20.7. This compares favourably with a historical price-earnings multiple for companies listed under the support services sector of the JSE of 7.4 times. It may be assumed that Park Ltd will accept this consideration and sell its 35.3% shareholding.

(iii) An investment bank has agreed in principle to fund R36m of the consideration with unsecured debt, to be repaid in equal annual installments over four years and bearing interest at 2% above the prime overdraft rate. In addition, the bank will provide R69m of secured debt to be repaid in equal annual installments over six years at 0.5% above the prime overdraft rate. The prime overdraft rate is currently 13%.

(iv) A new company, Newco (Pty) Ltd, will be incorporated with its shares held by management and the investment bank. Newco (Pty) Ltd will be funded by way of ordinary shares, Page 13-63 convertible preference shares, and the unsecured and secured loans. Management will acquire 10% of its issued share capital.

(v) Newco (Pty) Ltd will acquire the assets and liabilities of Jet Pack (Pty) Ltd.

(vi) The subsidiary's property has been professionally valued on the depreciated replacement cost basis at R89.7m. An outside party has agreed to purchase the property for this price, thereafter the company will be liquidated. Any profits generated by the company have traditionally been paid to the shareholders by way of interest on their loans. The company will repay the shareholder's loans, and the cash then remaining in the

company will be distributed to the shareholders by way of a liquidation dividend. Newco (Pty) Ltd will occupy the premises and has the option to buy the property after a period of ten years. This transaction will result in a rental cost in 20.8 of R10.08m, escalating at 15% per annum thereafter.

(vii) The corporate tax rate is 27%.

Required:

(a) Calculate the goodwill paid on the acquisition of the business of Jet Pack by Newco (Pty) Ltd.

(b) Calculate the funding requirements to purchase the interests of both shareholders of Jet Pack.

(c) Construct the abridged Statement of Financial Position (without notes thereto) of Newco (Pty) Ltd at 1 January 20.8 giving effect to the buy-out.

(d) State what additional matters should be considered by the management of Jet Pack before continuing with the proposed management buy-out of the business.

(*SAICA ITC/QE – adapted*)

Question 13.29

Almetals Ltd is considering the construction and operation of an aluminium smelter in Mozambique, situated approximately 20 km from Maputo harbour. An initial feasibility study has led to the conclusion that this project has a 'very good chance of being successful and profitable'.

In a post-commissioning audit of a similar investment project for an aluminium smelter near Richards Bay, the conclusion was that the Richards Bay project had proved to be very successful. There had been significant savings in budgeted costs and it had been completed three months ahead of schedule. The Richards Bay smelter is currently operating at an increase of 7% above the originally estimated capacity. The following excerpts from the Maputo project feasibility study are available:

1. Aluminium market and prices

Market analysts have forecast a significant increase in the per capita consumption of aluminium in developing Asian economies such as China, India and South Korea.

World aluminium production has increased substantially over the past few years as a result of improved efficiency, technological upgrades and the re-opening of idle capacity.

Most analysts forecast that anticipated demand will continue to exceed supply. It is estimated that world production capacity will have to increase by about 220 000 tons per year in order to satisfy future demand. An average LME (London Metal Exchange) price of US$1 800 per ton (in terms of December 20.1 values) is forecast for the next ten years.

2. Capital cost and construction

Based on the experience gained on the Richards Bay smelter project (with some minor adjustments for the Maputo smelter), the capital cost of the Maputo smelter is estimated at US$1 230 million. The cost will be evenly incurred over the two-year construction period.

Construction of the smelter, which will have a capacity of 340 000 tons per annum, is expected to commence in January 20.2 and will be undertaken by the same joint venture partners who were responsible for the construction of the Richards Bay smelter. Page 13-64

3. Capacity

The smelter will be in operation by the beginning of 20.4.

Capacity utilisation is planned as follows:

20.4 40%

20.5 60%

20.6 and later 100%

4. Operating costs

Operating costs comprise the following items (all amounts are constant 20.1 prices):

- *Alumina*. Alumina will be imported from Richards Bay, South Africa at a landed cost of US$340 per ton. One ton of alumina yields one ton of aluminium.
- *Electricity*. Electricity will be acquired from the Eskom network. Following initial discussions with Eskom, two lines will be provided for the plant. One line will run through Swaziland, while the other will run north of Swaziland. Either of the lines has the capacity to provide for the full electricity requirements of the smelter. Electricity costs in 20.4 are estimated at R8 000 000 and at R10 951 200 in 20.5.
- *Salaries*. Salaries will be paid in rand and are estimated at R5.2 million per year.

- *Other raw materials*. The costs are estimated at US$210 per ton of aluminium.
- *Overheads*. Overheads will amount to US$2.4 million per year.

5. Project financing

A decision has been taken to structure the project as a project financing transaction. The investment will thus be housed in a separate entity, managed by Almetals Ltd.

The following financing structure has been proposed:

	US$ Million
Equity	440
Quasi-equity	263
Debt	527
	1,230

* Interest bearing convertible shareholders loans

6. Discount rate

For the purposes of the feasibility study the discount rate was based on the following information: Cost of equity 14.5%

Cost of quasi-equity 13%

Cost of debt 10%

These rates represent United States dollar returns. The United States inflation rate is forecast at 3% and the interest rate differential between South Africa and the United States in expected to amount to 8%.

7. Taxation

The Mozambique government has indicated that the project will have to pay a levy of 0.5% on turnover as tax. No other taxation will be levied.

8. Terminal value

The terminal value of the project at the end of 20.13 is estimated at US$200 million.

9. Exchange rate

The exchange rate of the rand to the US dollar has been assumed to be R6.20: US$1.

A portion of the equity stake has been made available to outside venture capital investors, and Almetals Ltd has invited Equity Invest Ltd to make an investment. You

have been retained as a financial advisor to Equity Invest Ltd to assess the offer.

Required:

(a) Discuss the possible risks related to this project and indicate how these risks could be managed or hedged.

(b) Evaluate the proposed financing structure. Your evaluation should include the factors that would influence the maximum project debt/equity ratio that could be tolerated in the structure.

(c) Advise Equity Invest Ltd whether you would recommend an equity exposure to the project. List any additional information that you would deem necessary for purposes of your recommendation and test the project's sensitivity with respect to demand and the aluminium price.

(*SAICA ITC/QE – Financial Management Specialism*)

Question 13.30

Cloth Group Ltd ('Cloth Group') is a clothing retailer with 55 stores throughout South Africa. These stores, which focus on fashion clothing, are situated in suburban shopping malls and their target market is 25 to 45-year old females and males. Approximately 75% of sales are on credit and customers are granted 30 days' interest-free credit from the date of the statement. If payments are not made within this period, customers have to pay interest on the outstanding balance, calculated from the date of purchase, at the prime overdraft lending rate (currently 15.5% per annum) plus 4%. Customers are required to settle the full amount owing on a purchase within six months.

Cloth Group is listed on the JSE. The company has performed well over the past two years, reporting revenue of R412 million for the year ended 31 January 20.9 (20.8:R357 million) and achieving a gross profit margin of 42% for the 20.9 financial year (20.8:41%).

Cloth Group purchases clothing from foreign and local suppliers. Suppliers are selected based on the quality of merchandise supplied, pricing and reliability. The majority of foreign suppliers are based in Europe and invoice Cloth Group in euro (€). The audited statement of financial position of Cloth Group at 31 January 20.9, together with comparative figures, is set out on the following page: Page 13-66

Statement of Financial Position at 31 January		
	20.9	**20.8**
ASSETS	**R000**	**R000**
Non-current assets		

Statement of Financial Position at 31 January		
Property, plant and equipment	102,280	89,180
Current assets	198,050	169,800
Inventories	49,800	43,340
Trade receivables	129,600	110,180
Cash and cash equivalents	18,650	16,280
Total assets	300,330	258,980
EQUITY AND LIABILITIES		
Share capital	15,000	15,000
Retained earnings	192,875	153,000
Total equity	207,875	168,000
Current liabilities	92,455	90,980
Current borrowings (bank overdraft)	29,050	42,090
Trade and other payables	34,080	27,100
Shareholders for dividends	11,250	9,750
Taxation payable	18,075	12,040
Total equity and liabilities	300,330	258,980

RAISING OF MEDIUM-TERM FINANCE

The Treasury Division of Cloth Group has been tasked with raising finance of between R100 million and R125 million. Cloth Group requires the funding to grow its trade receivables and consequently revenue. The Treasury Division has received various formal proposals and has short-listed the following two proposals:

1 Euro denominated bond

Total nominal value	€ 9,600,000
Coupon(fixed)	6.9%
Issue date	31 March 20.9
Maturity date	31 March 20.12

Interest coupon on the bond is payable annually in arrears. The total nominal value is repayable in full on the maturity date.

2 Syndicated loan

Principal amount	R 120,000,000
Nominal interest rate	3 month JIBAR plus 2.5%
Loan advance date	31 March 20.9
Term	3 years
Upfront issuance cost	2% of the principal amount

Interest on the loan is to be calculated and paid quarterly in arrears. The current Johannesburg Interbank Agreed Rate (JIBAR) is 12.5%. The principal amount is to be

repaid in three equal annual instalments, with the first repayment due on 31 March 20.10. The upfront issuance cost is to Page 13-67 be paid by Cloth Group on the loan advance date, or the company can elect to receive 98% of the principal amount on the advance date.

Cloth Group's policy is to hedge all foreign currency exposure. Their commercial bankers have quoted the following spot and forward exchange contract rates:

Current spot rate	€1 :R12.50
12 months forward to 31 March 20.10	€1 :R13.75
24 months forward to 31 March 20.11	€1 : R15.10
36 months forward to 31 March 20.12	€1 :R16.60

Capital structure

Cloth Group has historically funded its growth out of its own cash flows and through short-term borrowings, mainly in the form of bank overdrafts. The Chief Executive Officer of Cloth Group has asked the Financial Director to determine what impact the raising of medium-term finance will have on the weighted average cost of capital (WACC) of Cloth Group. Information that may be relevant in determining Cloth Group's WACC is set out in the table below:

Number of shares in issue	15,000,000
Current share price	R 24.60
Beta co-efficient	0.90
Effective normal income tax rate	27%
Dividend per share declared on 31 January 20.9	R 0.75

Other information

Current yield of the R204 RSA government bond, maturity date 21 December 20.18	9.0%
Current yield of the Rl 53 RSA government bond, maturity date 31 August 20.10	9.4%
Premium for market risk	8.0%

Required:

(a) Calculate and determine which debt instrument (euro denominated bond or the syndicated loan) will be the most cost effective way for Cloth Group Ltd to raise medium-term finance. Ignore all tax implications.

(b) Identify and describe the key factors that Cloth Group Ltd should consider in evaluating which debt instrument to issue.

(c) Calculate, with reasons, the weighted average cost of capital (WACC) of Cloth Group Ltd at 31 March 20.9, assuming that the company elects to issue the euro denominated bond.

(d) Assuming that Cloth Group Ltd elected to enter into the syndicated loan agreement, discuss the key factors and issues that Cloth Group Ltd should consider in evaluating whether to change from a floating interest rate to a fixed interest rate.

(e) Briefly describe the concept of asset-backed securitisation and indicate what benefits, if any, could accrue to Cloth Group Ltd from securitising its trade receivables.

(f) Identify the key procedures that Cloth Group Ltd should follow in assessing the creditworthiness of new customers.

(*SAICA ITC/QE*)

Question 13.31

ElectriBolt Ltd ('ElectriBolt') is an independent electricity supplier with various power-generation operations throughout South Africa. ElectriBolt is listed on the main board of the JSE. The company's most recent financial reporting date was 31 December 20.9. Page 13-68

Background

On 1 January 19.9, ElectriBolt entered into a unique hydro-electricity supply licence agreement with the South African government ('the government'). The licence agreement entitled ElectriBolt to install three turbines at the Augrabies Waterfall, which is situated in the Augrabies National Park, and generate and supply electricity for a period of 15 years. The licence agreement provides that:

- ElectriBolt is required to pay the government a fixed instalment of R800 000 annually in arrear for the right to use the site to generate electricity; and
- the agreement is not renewable at the end of the 15-year term.

ElectriBolt decided to grant the right of use of the abovementioned supply licence, from the inception date of the agreement with the government (1 January 19.9), to PowerSmart Ltd ('PowerSmart'), a large electricity supplier, for a period of 15 years. All licensing rights granted to ElectriBolt by the government were transferred to PowerSmart in exchange for a fixed annual instalment of R1 million, payable in arrear to ElectriBolt. By law the final operator of an electricity supply business is solely responsible and liable for any related environmental site rehabilitation. PowerSmart may not transfer or sell the supply licence to any other party. A cancellation penalty of R900 000 is payable by PowerSmart to ElectriBolt should PowerSmart at any stage unilaterally decide on the early cancellation of the agreement. On 1 January 19.9, ElectriBolt did not recognise any assets or liabilities in respect of the hydroelectricity

supply licence with the government and the right of use of the licence granted to PowerSmart. The R800 000 per annum for the supply licence is expensed on an annual basis and the R1 million per annum receipt from PowerSmart is recognised as revenue on an annual basis. This accounting policy is acceptable in terms of International Financial Reporting Standards.

Feasibility study on PowerSmart's Augrabies operations

ElectriBolt recently established that PowerSmart is underperforming at the Augrabies site and believes that significantly more electricity can be generated during the last four years of the supply licence. The major reason for PowerSmart's disappointing performance is the regular labour disputes experienced at the Augrabies operation. ElectriBolt conducted a feasibility study in December 20.9 to evaluate the possible acquisition of PowerSmart's Augrabies operations. Following this feasibility study, ElectriBolt proposed acquiring the PowerSmart Augrabies division as a going concern, including all assets and liabilities of this division except for cash and cash equivalents and taxation liabilities. The licensing agreement between ElectriBolt and PowerSmart would be terminated as part of the acquisition. The Chief Financial Officer (CFO) of ElectriBolt prepared the following cash-flow projections, following a detailed review of historic financial information of the Augrabies division of PowerSmart and its budgets for the next four years, for the purposes of valuing the Augrabies division:

Year ending 31 December	Notes	20.10	20.11	20.12	20.13
Turnover	1	18,000,000	18,000,000	18,000,000	18,000,000
Operating costs	1	-4,500,000	-4,500,000	-4,500,000	-4,500,000
Opportunity cost	2	-1,000,000	-1,000,000	-1,000,000	-1,000,000
Supply licence agreement instalments		-800,000	-800,000	-800,000	-800,000
Operating cost savings	3	300,000	300,000	300,000	300,000
Cost of helicopter lease	4	-480,000	-480,000	-480,000	-480,000
Depreciation: Turbines		-700,000	-700,000	-700,000	-700,000
Interest on long-term loan	5	-897,536	-712,469	-503,344	-267,032
Environmental site rehabilitation costs	6	-	-	-	-2,500,000
Pending legal claim	7	-	-	-	-
Taxation	8	-2,679,065	-2,729,033	-2,785,497	-2,174,301
Net cash flows		7,243,399	7,378,498	7,531,159	5,878,667
Related calculations				**Notes**	
Weighted average cost of capital (WACC)				9	23%
NPV of the future cash flows for the Augrabies division of PowerSmart	9	17,381,497			

Notes to the cash-flow projections

(1) Page 13-69 Projected turnover includes a conservative estimate of the additional electricity that ElectriBolt could generate and supply, assuming it acquired control of the operations. Operating costs exclude annual supply licence payments due by PowerSmart to ElectriBolt.

(2) Opportunity cost represents annual instalments in terms of the supply licence agreement, to which ElectriBolt will no longer be entitled.

(3) PowerSmart incurred research and development costs during the period 20.7 to 20.9 aimed at improving operating efficiency. As a result thereof, PowerSmart expects a possible annual operating cost saving of R300 000 from 20.10 to 20.13. The CFO of PowerSmart is of the opinion that the cost saving is only 45% probable, resulting in the development costs not meeting the recognition criteria in terms of IAS 38, *Intangible Assets*, for recognition as an intangible asset. It follows that development costs were immediately expensed when incurred.

(4) PowerSmart leases an executive helicopter from Fly-with-Me Airways (Pty) Ltd at a fixed instalment of R360 000, payable annually in arrear. The lease agreement was correctly classified as an operating lease in terms of IAS 17, *Leases*. The lease agreement commenced on 1 January 20.8 and ends on 31 December 20.12. On 31 December 20.9, it was reliably established that similar executive helicopters can be leased at a market related fixed instalment of R480 000, payable annually in arrear.

(5) PowerSmart obtained a long-term loan of R15 million on 1 January 19.9 to finance this particular operation. The long-term loan is repayable in 15 equal annual instalments, which commenced on 31 December 19.9. The loan bears interest at a fixed rate of 13% per annum. The loan agreement provides that the loan cannot be repaid earlier than the agreed repayment profile. On 31 December 20.9, long-term loans with a similar risk profile and remaining maturity were available at a fixed rate of 12% per annum.

(6) This amount has been reliably estimated by an independent environmental rehabilitation expert.

(7) Labour unrest increased after the recent dismissal of a number of PowerSmart's employees as a result of increased operational inefficiency. The trade union to which these employees belong has instituted a legal claim against PowerSmart on behalf of the employees on the grounds of unfair dismissal. The legal advisers of PowerSmart are of the opinion that the dismissed employees have a valid claim against PowerSmart for unfair dismissal; the trade union will not be able to prove the claim for unfair dismissal in court, due to a lack of evidence; and it is possible but not probable that the court will require PowerSmart to make a financial settlement to the dismissed employees. The CFO did not include any amount relating to the legal claim in the forecast cash flows. Should such a claim be successful, any amount paid by PowerSmart will not be deductible for tax purposes. The fair value of the

legal claim at 31 December 20.9, as reliably determined by an experienced actuary, is R450 000.

(8) All items in the cash-flow budget have been assumed to be taxable or deductible for income tax purposes, except as per point 7 above.

(9) The nominal WACC of ElectriBolt is 23% and the forecast cash flows have been discounted using this rate.

Financing alternatives

On 31 December 20.9 the Augrabies division of PowerSmart was acquired by ElectriBolt as a going concern, including all assets and liabilities of the division except for cash and cash equivalents and taxation liabilities. The purchase consideration of R16 million was paid by ElectriBolt on 31 December 20.9. It was correctly established that the transaction between ElectriBolt and PowerSmart constitutes a 'business combination' as defined in IFRS 3, *Business Combinations.* ElectriBolt is considering various financing alternatives for the business combination transaction:

- Payment out of existing cash reserves of R16 million; or
- Obtaining a R16 million medium-term loan from ElectriBolt's bankers. The loan is to bear interest at 1% above the prevailing prime overdraft rate. This is the company's incremental Page 13-70 cost of borrowing. The loan is to be repaid in one bullet payment at the end of four years. Interest is to be calculated and compounded annually in arrear, and capitalised into the outstanding loan balance. Transaction costs of 1% of the principal amount will be payable at the inception of the medium-term loan. The interest to be incurred on such a long-term loan is deductible for taxation purposes in terms of Section 24J of the Income Tax Act; or
- The issue of compulsory convertible preference shares with a par value of R16 million. Preference shareholders will be entitled to an annual dividend calculated as 80% of the prevailing prime overdraft rate multiplied by the par value of shares held. ElectriBolt is required to pay preference dividends annually in arrear and has no discretion with regard to declaring these dividends. Each preference share will automatically convert into one ordinary share after four years. Analysts predict that the value of the converted shares at the end of year four will amount to R17 800 000.

The information below represents an extract from the **draft** statement of financial position of the Augrabies division of PowerSmart as at 31 December 20.9:

Draft statement of financial position of the Augrabies division of PowerSmart as at 31 December 20.9

	Notes	R
ASSETS	a	9,750,000
Non-current assets		
Property, plant and equipment		
Current assets	b	750,000
Inventories		
Trade receivables	c	300,000
Cash and cash equivalents		250,000
LIABILITIES	d	5,480,535
Non-current liabilities		
Long-term borrowings		
Deferred tax	e	571,050
Long-term provisions	f	-
Current liabilities	g	435,000
Trade payables		
Current portion of long-term borrowings	d	1,423,590
SA Revenue Service		115,000

Notes

(a) Items of property, plant and equipment are subsequently measured according to the cost model in terms of IAS 16, *Property, Plant and Equipment*. The market value of the property, plant and equipment as at 31 December 20.9 was R14 million.

(b) The fair value of inventories was reliably determined at R800 000 as at 31 December 20.9.

(c) The balance of net trade receivables comprised the following as at 31 December 20.9:

	R
Gross trade receivables	480,000
Less: Allowance for doubtful debts	-180,000
	300,000

SARS grants a deduction of 25% of the allowance for doubtful debts for taxation purposes. The fair value of trade receivables as at 31 December 20.9 was R400 000.

(d) Long-term borrowings are subsequently measured according to the amortised cost model in terms of IAS 39, *Financial Instruments: Recognition and Measurement*. Long-term borrowings consisted of the following as at 31 December 20.9:

	R
Long-term loan	6,904,125
Less: Current portion of long-term borrowings	-1,423,590
	5,480,535

(e) The deferred tax balance as at 31 December 20.9 was calculated as follows:

Asset/liability	Carrying amount	Tax base	Temporary difference
Property, plant and equipment	R 9,750,000	R 7,500,000	R 2,250,000
Inventories	750,000	750,000	0
Trade receivables	300,000	435,000	-135,000
Long-term loan	6,904,125	6,904,125	0
Trade payables	435,000	435,000	0
Taxable temporary differences			2,115,000
Deferred tax calculated at 27%		27%	571,050

(f) No provision has been recognised in the statement of financial position of PowerSmart in respect of the environmental site rehabilitation. 70% of the damage caused to the environment occurred when the turbines were installed, while the remaining 30% of the damage to the environment is caused evenly over the duration of the contract. No environmental rehabilitation activities had been undertaken by 31 December 20.9 and the CFO therefore holds the opinion that no provision should be recognised in the statement of financial position until the electricity supply operation ceases. The SARS will allow the amount incurred in respect of environmental rehabilitation costs as a deduction for taxation purposes when it is actually paid.

(g) The fair value of trade payables as at 31 December 20.9 was reliably determined at R500 000.

(h) The supply licence, the use of which was granted by ElectriBolt, had a fair value of R4 500 000 at 31 December 20.9 based on the terms of the licensing agreement between ElectriBolt and PowerSmart. If a similar right with payments at market rates were granted at 31 December 20.9, it would have a fair value of R5 million.

Additional information

Where appropriate and unless stated otherwise, the SARS accepts the acquisition date fair values of assets and liabilities for taxation purposes. The current prime overdraft rate is 10% per annum.

Required:

(a) Identify, with reasons, any errors in and omissions from the cash flow forecasts and discounted future cash flows of the Augrabies division of PowerSmart Ltd as prepared by the CFO of ElectriBolt Ltd.

(b) Identify and describe any advantages and disadvantages of ElectriBolt Ltd settling the purchase consideration due to PowerSmart Ltd using its own cash resources.

(c) With regard to ElectriBolt Ltd evaluating the financing of the acquisition of the Augrabies division of PowerSmart Ltd through obtaining the medium-term loan or through the issue of the preference shares: (i) calculate and determine which instrument will be more cost effective for ElectriBolt Ltd to use; and (ii) discuss any other factors ElectriBolt Ltd should consider in deciding which instrument to use.

(*SAICAITC/QE - financial management section*) Page 13-72

Question 13.32

Build Ltd is a JSE-listed retailer of building materials and associated products that it sells directly to its customers through its stores in Southern Africa. The management of Build Ltd plans to expand into additional segments of the building supplies market. Build wishes to bid for 100% of the equity of NBS (Pty) Ltd which is a building materials wholesaler that provides building materials to the commercial and industrial markets. NBS has experienced a surge in growth and management wish to increase capital expenditure and open new stores during the next four years. NBS is owned by a large conglomerate that is seeking to divest from the building materials sector.

Build intends to fund the largest part of the acquisition of NBS with debt financing. The following alternatives have been identified:

1. R2 billion loan from Tselane Ventures (debt fund)

Interest	At the interbank one-year rate (currently 5%) plus 4% Compounded annually in arrears Capitalised to the outstanding loan balance
Upfront transaction costs	3% of the principal amount
Principal amount	Repayable in a single payment at the end of five years

2. R2 billion loan from Ace Bank

Interest	Fixed interest rate of 9% per annum Calculated and paid annually in arrears
Upfront transaction costs	Arrangement fee of 1% of the principal amount
Principal amount	Repayable in five equal annual capital payments at the end of each year

You have been tasked to analyse the financing options whilst a colleague has been tasked to analyse a preliminary valuation of NBS (see Question 6.40).

- Assume a company tax rate of 27%.
- All initial transaction and financing costs are tax deductible.

- Build generates sufficient other income to utilise any tax losses in the year that these are incurred.
- Tax payments or allowances are made in the same year as the relevant cash flow that created the tax liability/saving.

Required :

Calculate and conclude which of the two financing alternatives would be the most cost effective way for Build to raise the required financing to acquire NBS.

(*SAICAITC 2022 – adapted. This only includes the financing section. Refer to Q6.40 for the valuation section*)

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[1] Edcon's highly levered structure combined with poor operating returns and a high initial price led to the failure of this private equity investment for Bain Capital.

[2] In an amendment to the Pensions Funds Act, the allocation limit to PE has increased from 10% to 15%. This should lead to an increase in the allocation of funds by pension funds to private equity.

[3] In terms of section 240 of the Income Tax Act, a company may be able to deduct interest with regard to the purchase of shares if the company acquires a controlling equity interest of at least 70% in a target company, which is an operating company (i.e. a company that provides goods or services), but this deduction cannot exceed a specified percentage (about 40%) of adjusted taxable income.

[4] In terms of the Companies Act, 2008, a company may be required to apply the Solvency and Liquidity tests prior to redemption.

[5] The business rescue provisions of Chapter 6 of the Companies Act, 2008 is expected to reduce the effectiveness of security on loans.

[6] The cumulative inflation factor would be 117.978% (112.36% × 1.05), thereby resulting in a redemption value of R1 179 780 (R1m × 117.978%), and a nominal interest payment of R23 596 (R20 000 × 117.978%).

[7] However, section 23N limits any interest deduction.

[8] In terms of section 23N (effective from 1 April 2014 and which applies to a s45 intragroup transfer of assets and liabilities), any interest deduction is limited to 40% of adjusted taxable income (taxable income less interest income and recoupments and plus interest paid and depreciation deductions – essentially EBITDA from a tax perspective). This limit changed in 2015 to 40% × (repo rate 1 4%)/10%, subject to a maximum of 60%. If the repo rate is 5.5%, then the deduction will be limited to 38% (0.40 × (5.514)/10).

[9] Section 23M may also apply which regulates interest deductions where a debtor and a creditor are in a controlling relationship. Sections 23N, 23M and 240 are complex sections and

will probably change over time. It is important to obtain expert advice as these sections may affect financing transactions between connected parties and where there is a controlling interest.